DISCRETE MATH · PROPERTIES OF RELATIONS

Relation Property Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Relation Property Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. On A = {1, 2, 3}, is R = {(1, 1), (2, 2), (3, 3), (1, 2)} reflexive? Symmetric? Transitive?

2. How would you see "symmetric" in a directed graph?

2 Explore use the tool

1. Load "a ≤ b" on {1, 2, 3, 4}. Which properties hold?

2. Build a relation that is reflexive and symmetric but not transitive. Record it and the counterexample.

3. Make the chain (1, 2), (2, 3), (3, 4) and press Transitive closure. Which pairs were added?

4. On {1, …, 5}, load "a ≡ b (mod 3)". List the equivalence classes.

3 Record use the tool and fill in the table

Properties of relations on {1, 2, 3, 4}

Use "Load a relation" or build each one.

Relation	Reflexive	Symmetric	Antisymmetric	Transitive
a = b				
a < b				
a divides b				
same parity				
{(1, 2), (2, 1)}				

4 Explain what did you discover?

1. Reflexive: a _____ at every vertex. Symmetric: every arrow has a _____ arrow.

2. Transitive: every two-step path a → b → c has a _____ a → c.

3. An equivalence relation is reflexive, _____ and transitive; its classes form a _____ of A.

4. A student says “symmetric and transitive implies reflexive: from (a, b) and (b, a) we get (a, a).” Find the flaw and give a counterexample.

5 Apply a new situation, no tool

1. On the integers, is R = {(m, n) : m − n is even} an equivalence relation? Describe the classes.

2. How many ordered pairs are in the equivalence relation for the partition {{1, 2, 3}, {4, 5}}?

3. Find the transitive closure of {(1, 2), (2, 3), (3, 1)} on {1, 2, 3}.

4. Is “is a sibling of” transitive? Is it reflexive?